ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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The Wilson Loop Method and Numerical $\mathbb{Z}_2$ Classification

I. Coding for Spin Texture and Wilson Loop for Graphene with Rashba and Dresselhaus Type SOC Cases

II. Coding for Plotting Spin Texture and Wilson Loop

I. Coding for Spin Texture and Wilson Loop for Graphene with Rashba and Dresselhaus Type SOC Cases

Input File

```
2.46
Lattice Constant
1.0 0.0 0.0
-0.5 0.8660254037844386 0.0
0.0 0.0 1.0
C
2
Direct
0.6666666666 0.3333333333 0.0
0.3333333333 0.6666666666 0.0
onsite: 0.0 0.0
k point mesh for DOS: 500 500
Energy range for DOS: -5.0 5.0
Energy point for DOS: 1000
Broadening value for DOS: 0.05
Centre of k circle mesh: 0.3333333333 0.3333333333 0.0
Number of k circle mesh: 5
Delta radius of k circle mesh: 0.02
Delta angle of k circle mesh: 60.0 30.0 15.0 10.0 8.0
Intrinsic Spin Orbit Coupling strength: 0.005
Rashba SOC strength: 0.004
Dresselhaus SOC strength: 0.006
k point mesh for wilson loop: 500 500
```

Topological Non-Trivial Case

$$t_{rashba_soc} < 3\sqrt{3}t_{intrinsic_soc}$$

$$t_{dresselhaus_soc} < 3\sqrt{3}t_{intrinsic_soc}$$

I. Coding for Spin Texture and Wilson Loop for Graphene with Rashba and Dresselhaus Type SOC Cases

Input File

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Lattice Constant
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Centre of k circle mesh: 0.3333333333 0.3333333333 0.0
Number of k circle mesh: 5
Delta radius of k circle mesh: 0.02
Delta angle of k circle mesh: 60.0 30.0 15.0 10.0 8.0
Intrinsic Spin Orbit Coupling strength: 0.005
Rashba SOC strength: 0.04
Dresselhaus SOC strength: 0.04
k point mesh for wilson loop: 500 500
```

Topological Trivial Case

$$t_{rashba_soc} \geq 3\sqrt{3}t_{intrinsic_soc}$$

$$t_{dresselhaus_soc} \geq 3\sqrt{3}t_{intrinsic_soc}$$

I. Coding for Spin Texture and Wilson Loop for Graphene with Rashba and Dresselhaus Type SOC Cases

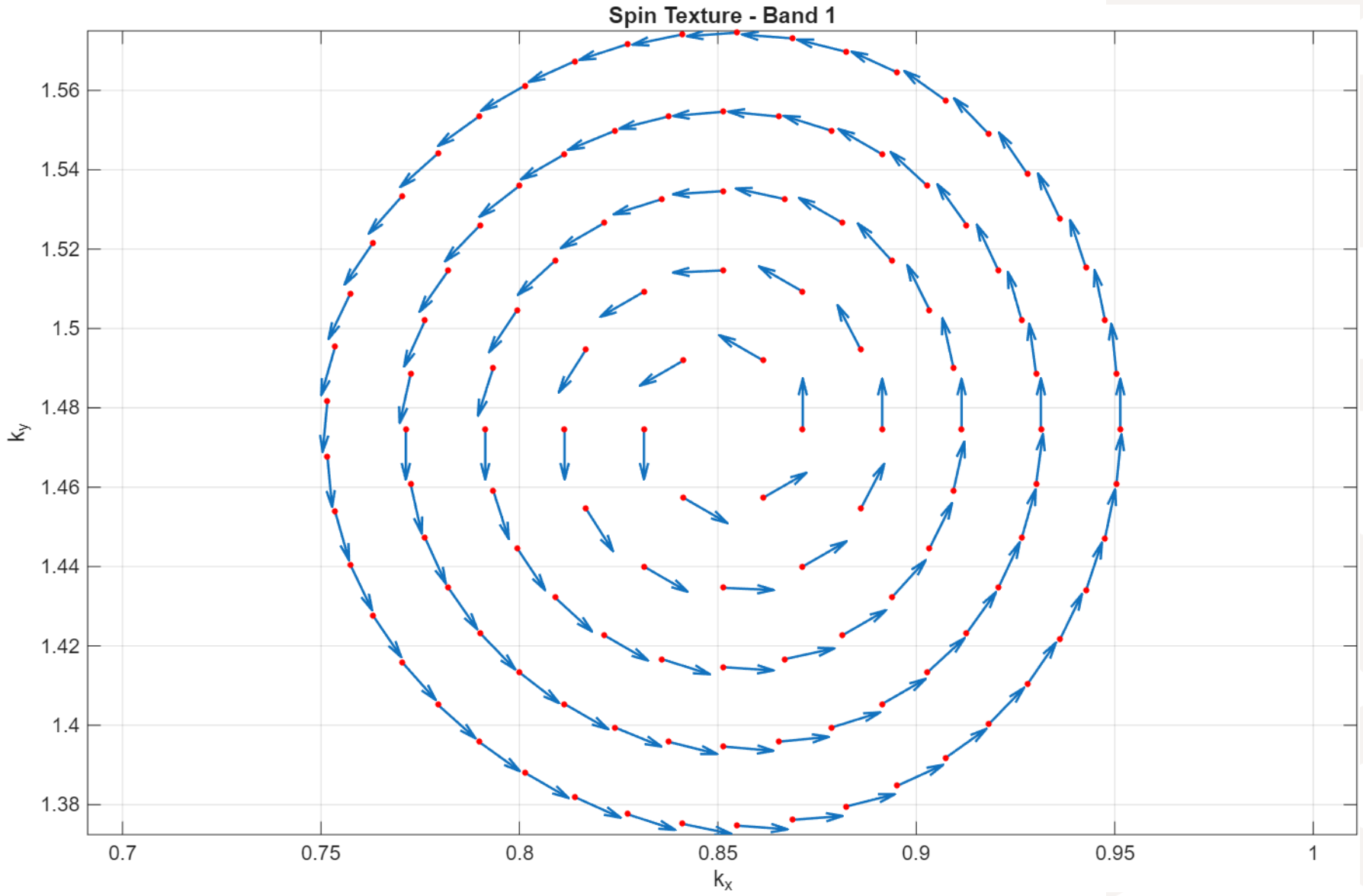
Please refer to the coding

II. Coding for Plotting Spin Texture and Wilson Loop

Please refer to the coding

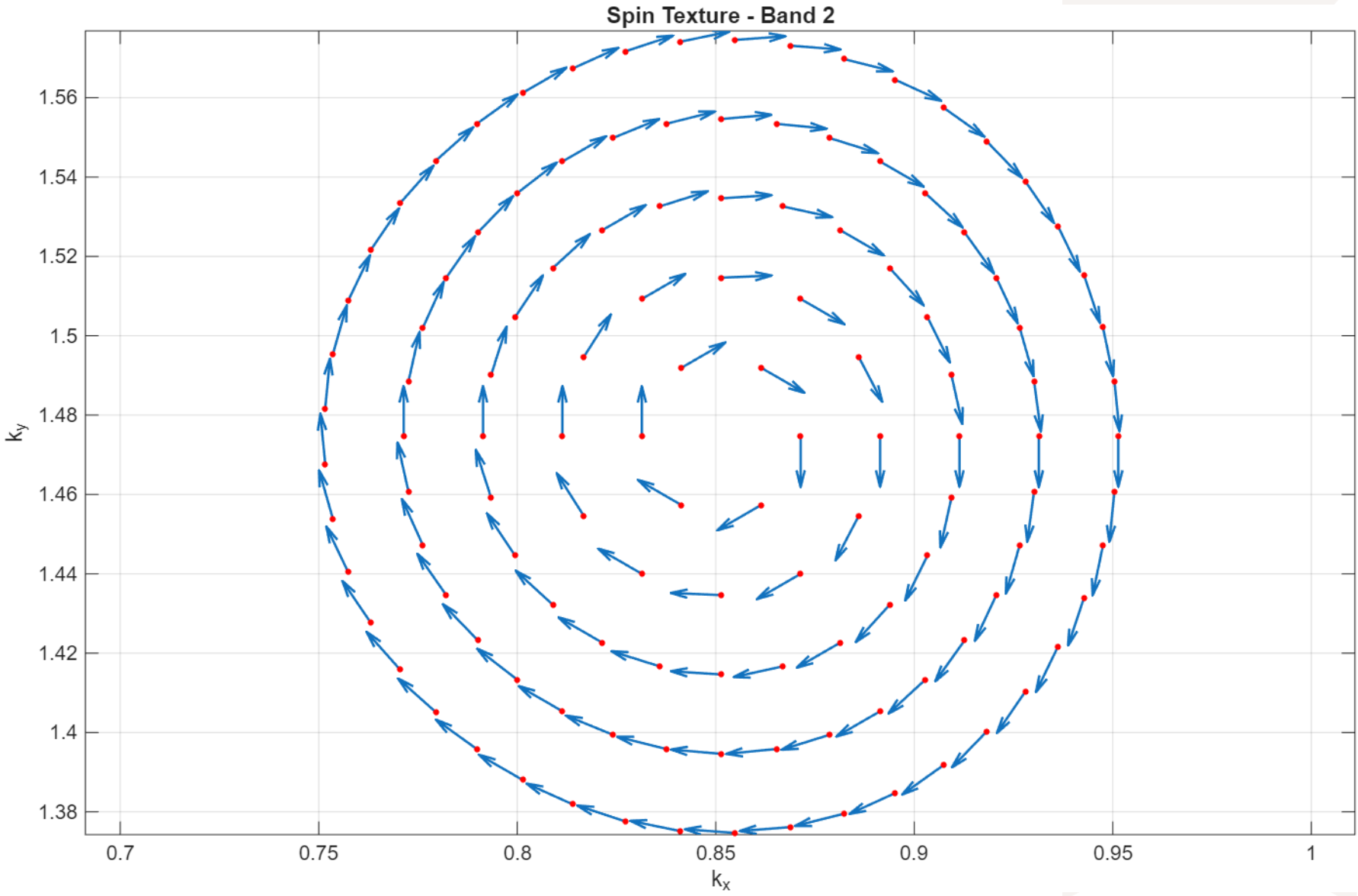
II. Coding for Plotting Spin Texture and Wilson Loop

Topological Non-Trivial Case



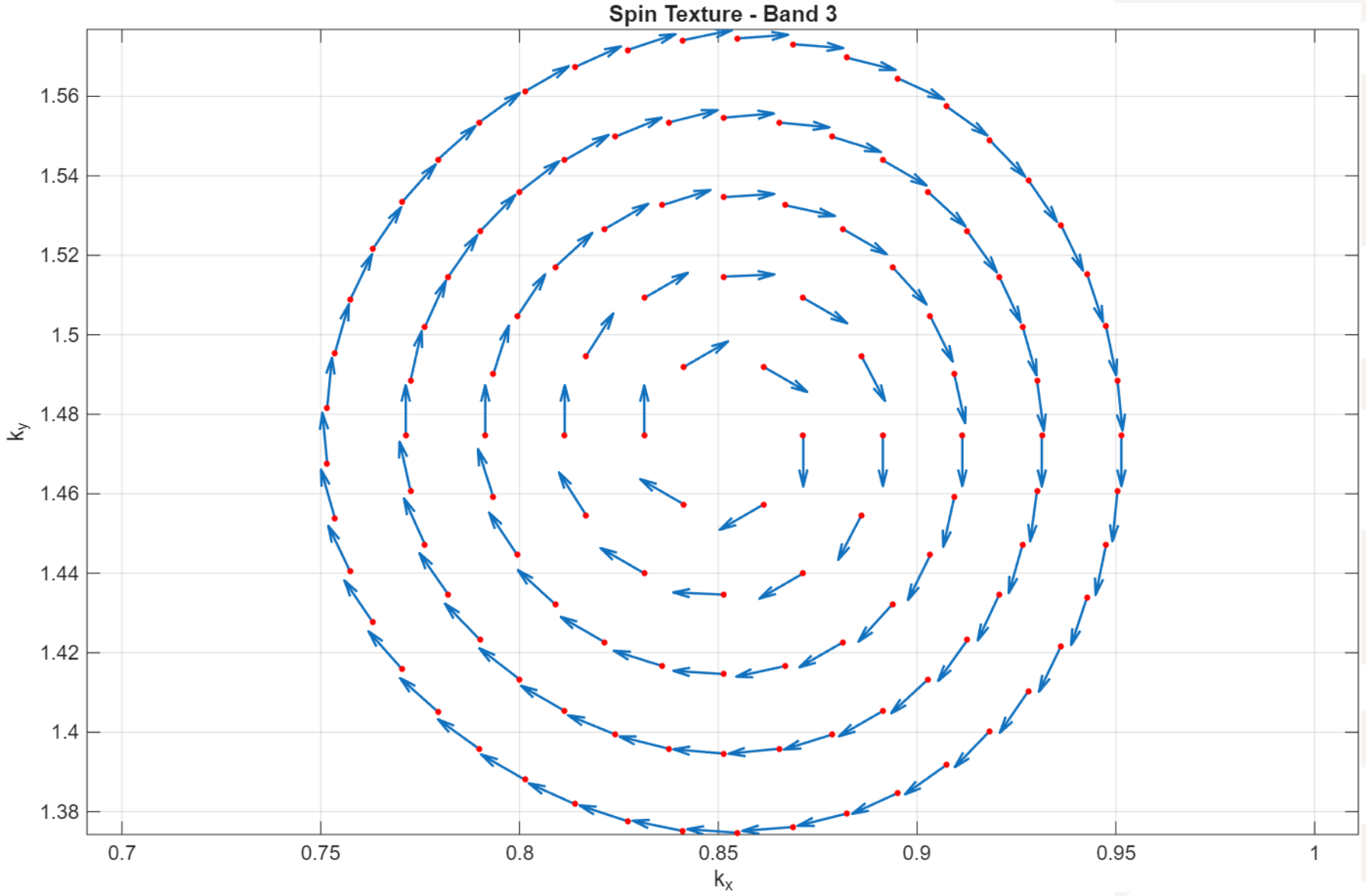
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Topological Non-Trivial Case



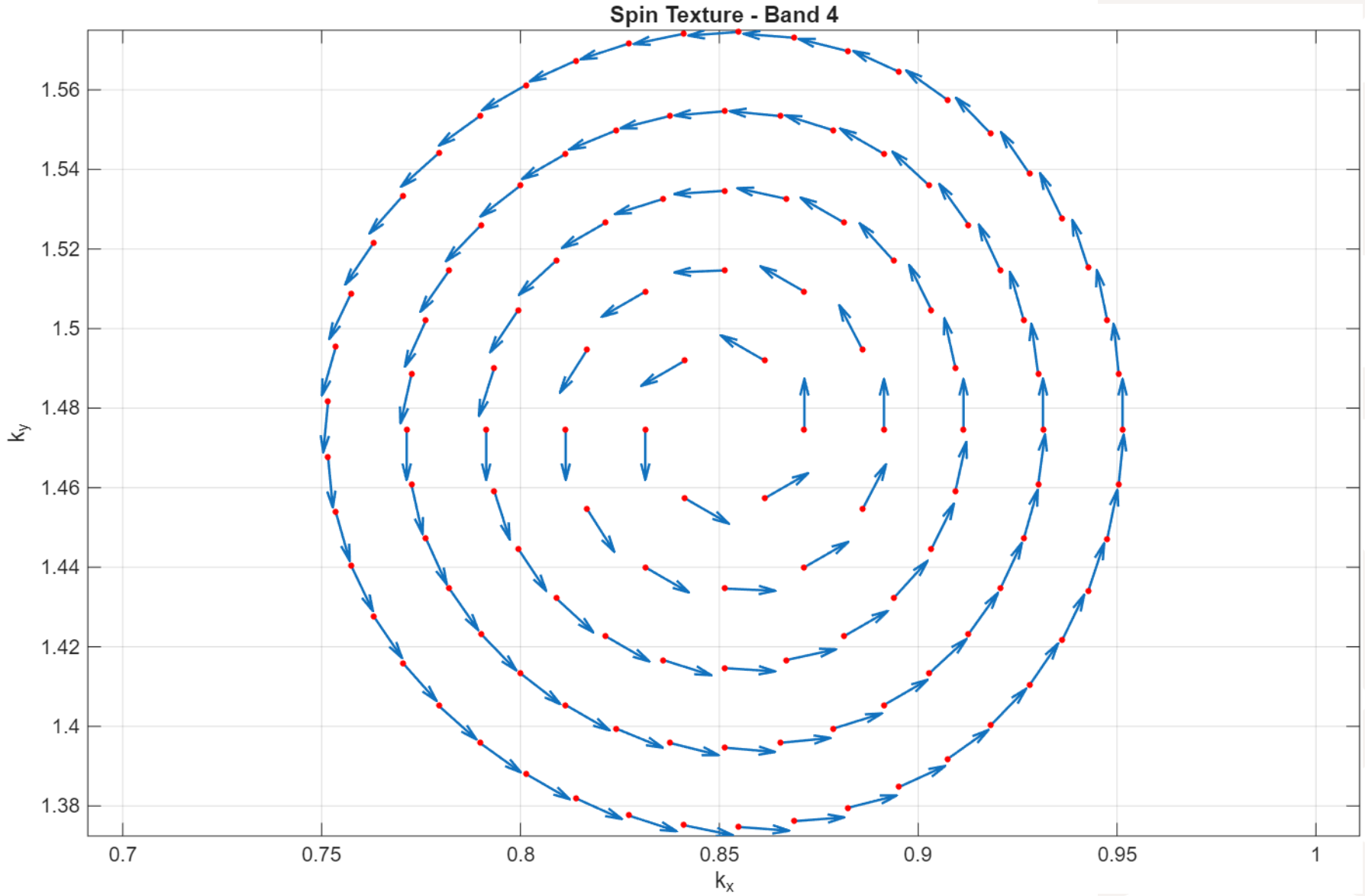
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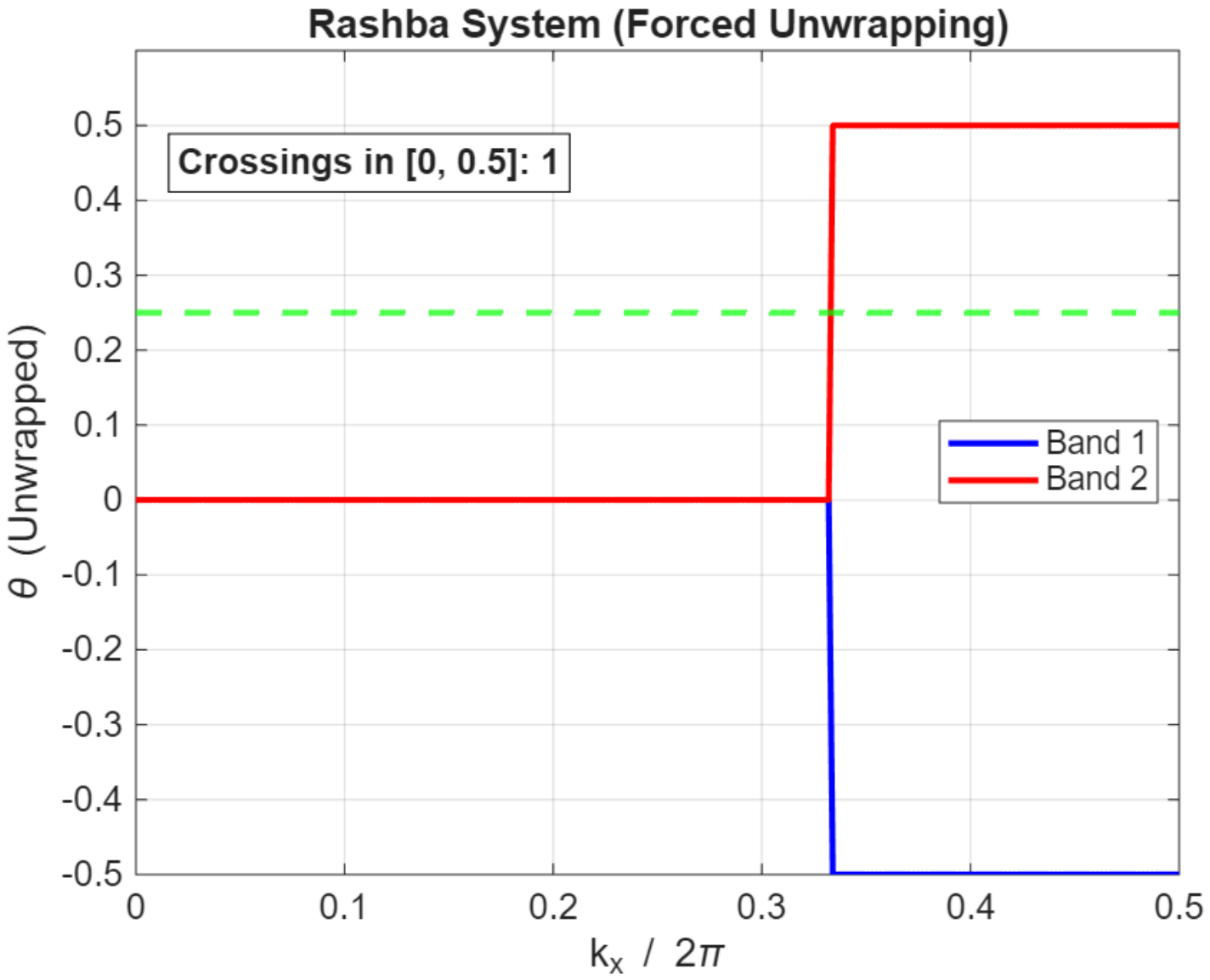
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Topological Non-Trivial Case



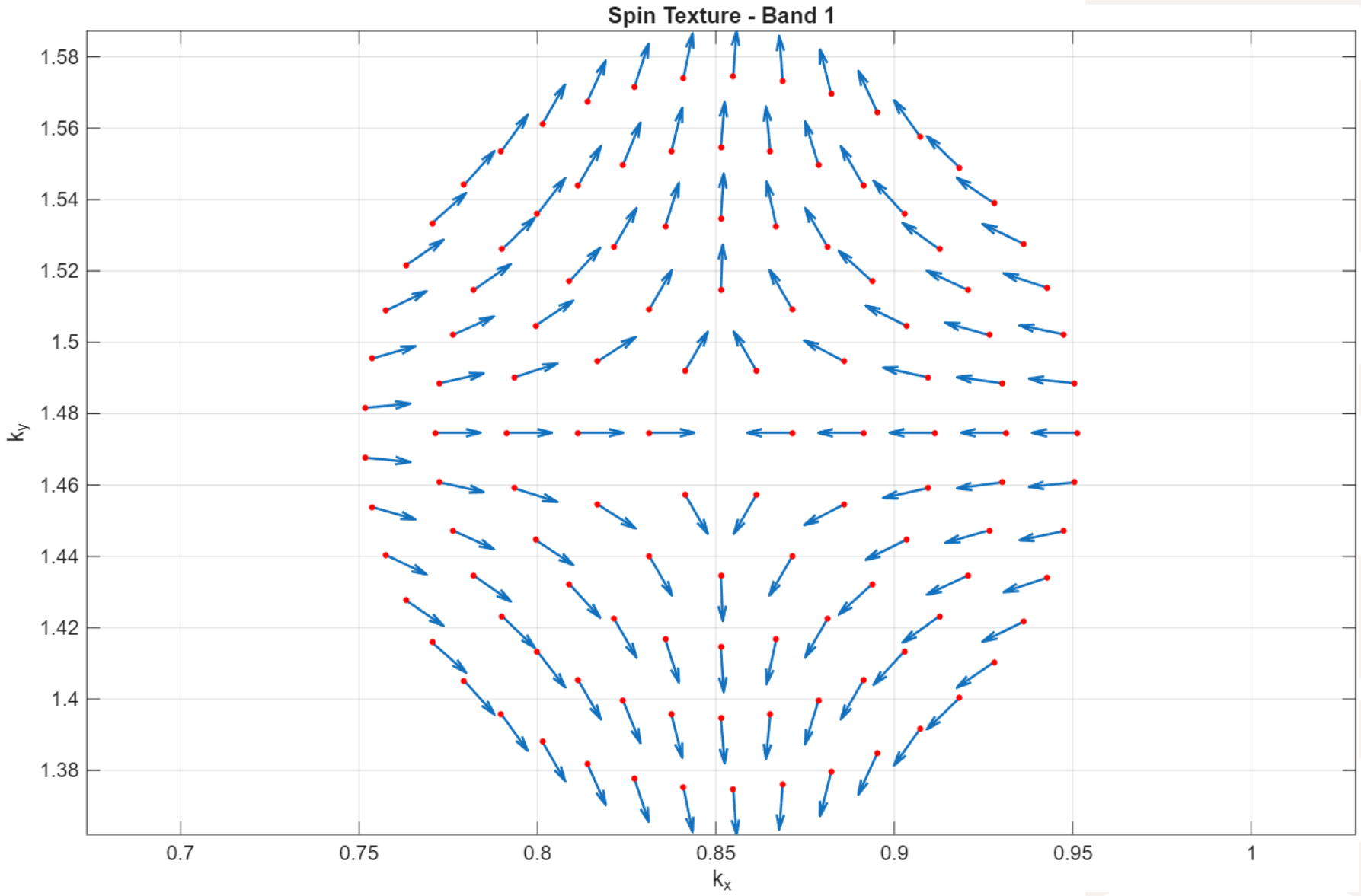
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Topological Non-Trivial Case



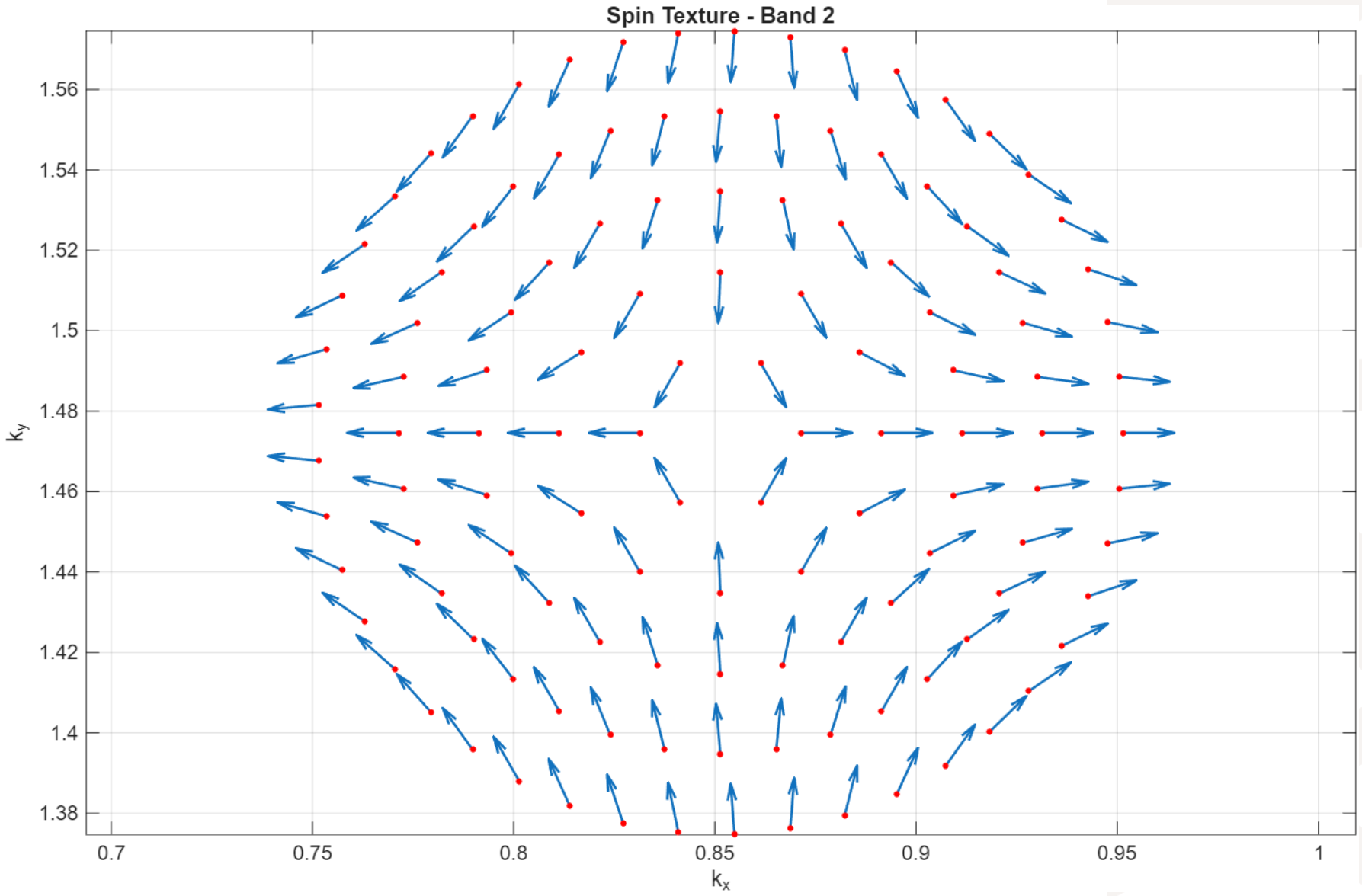
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Topological Non-Trivial Case



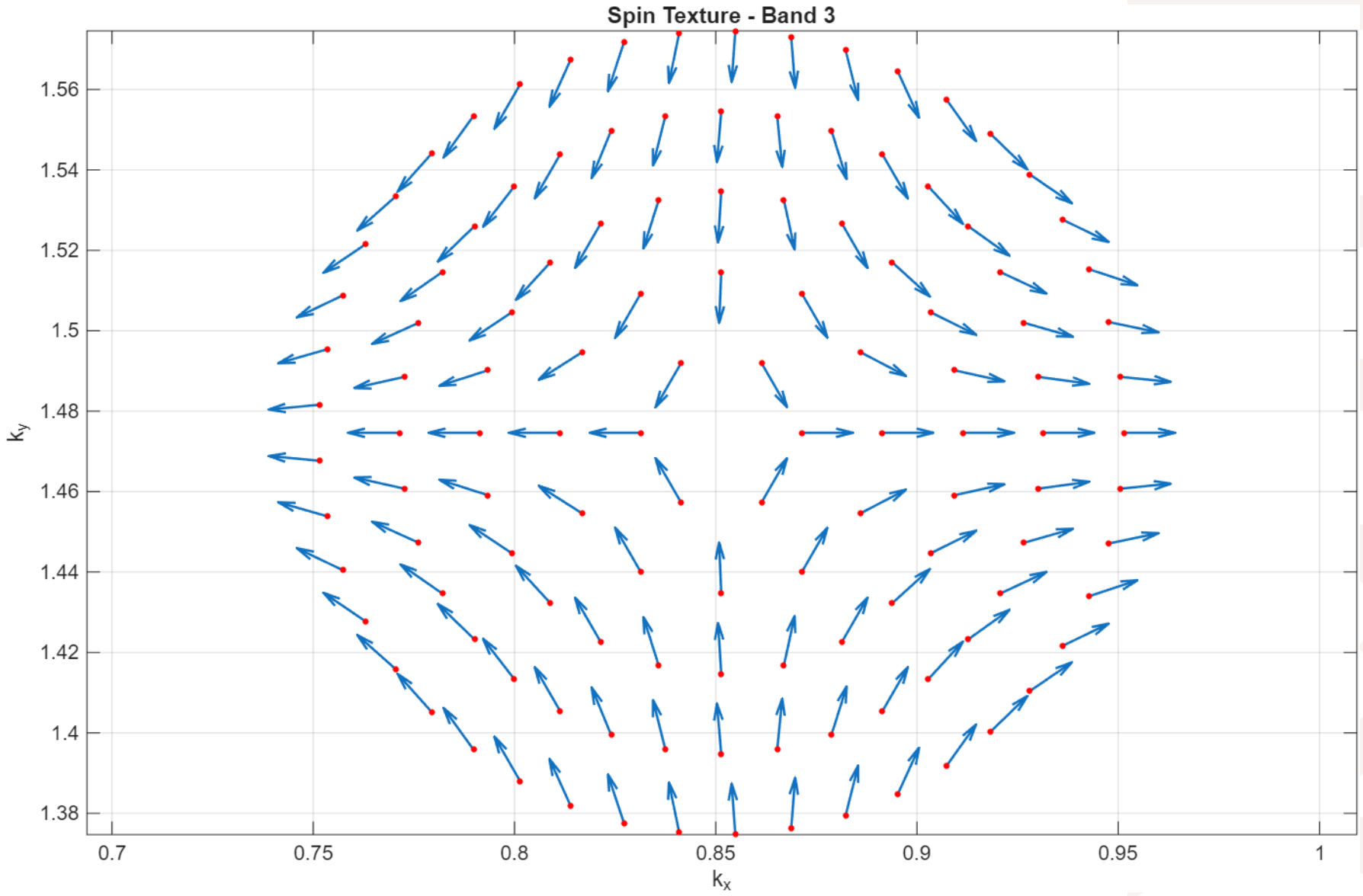
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Topological Non-Trivial Case



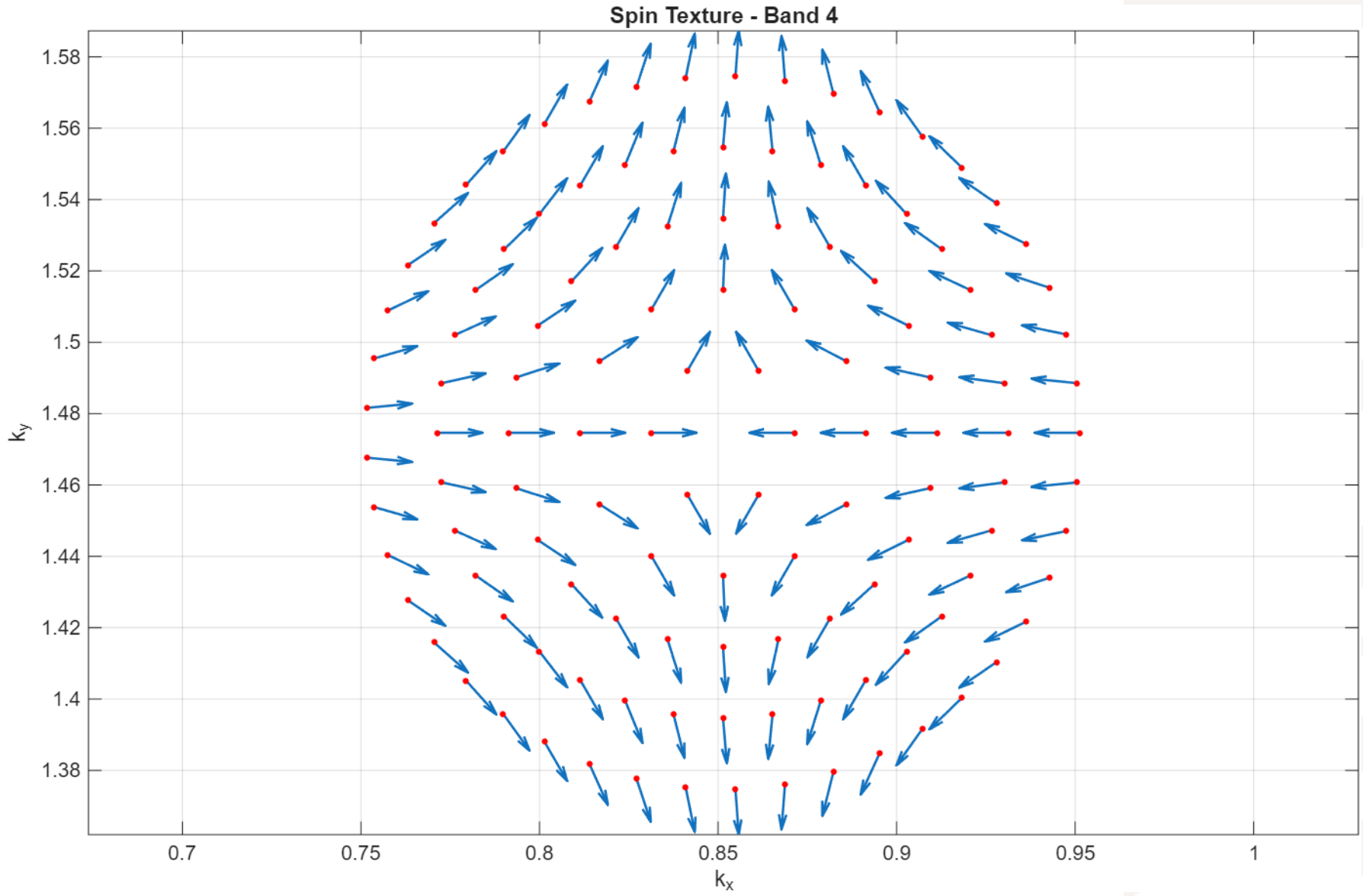
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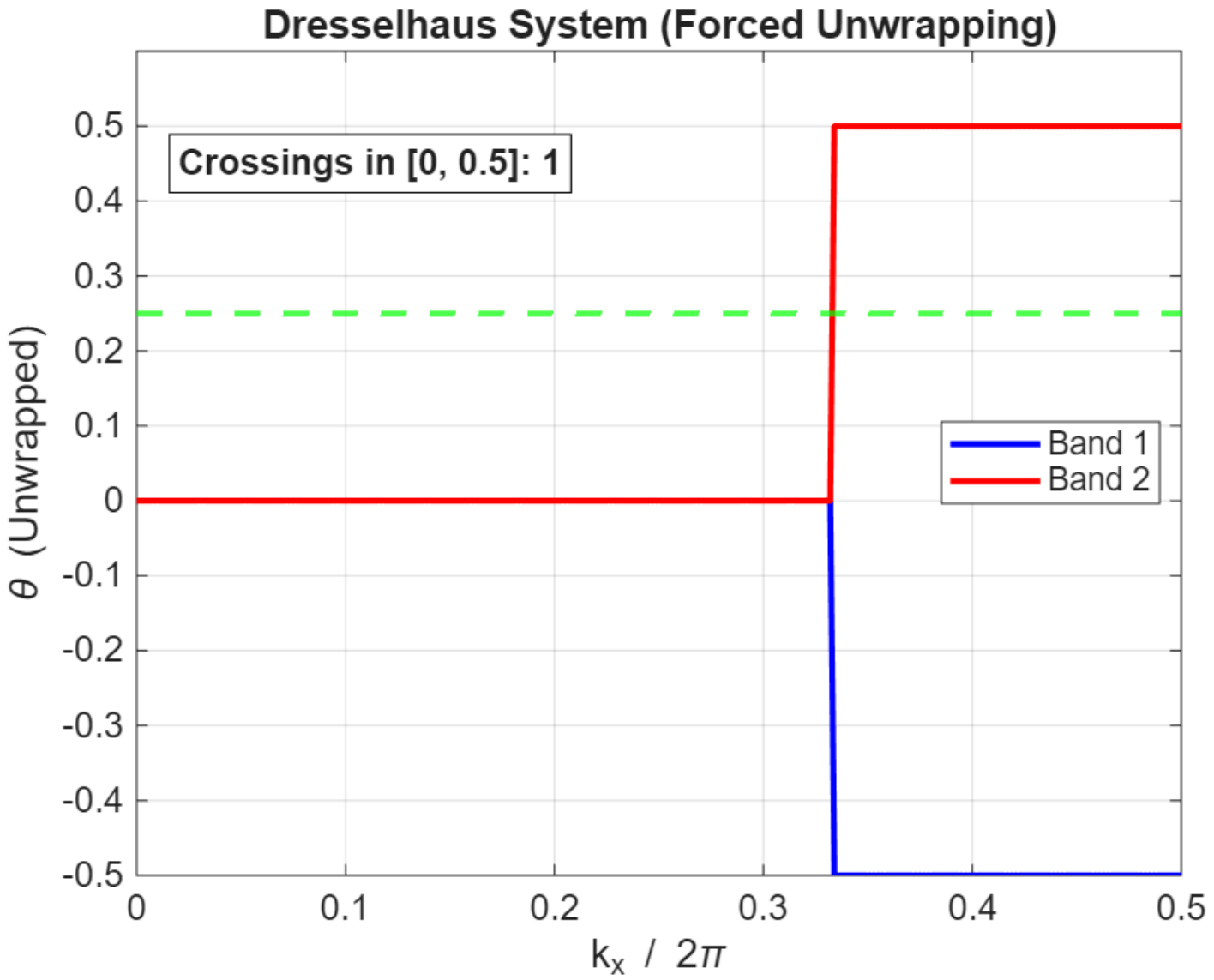
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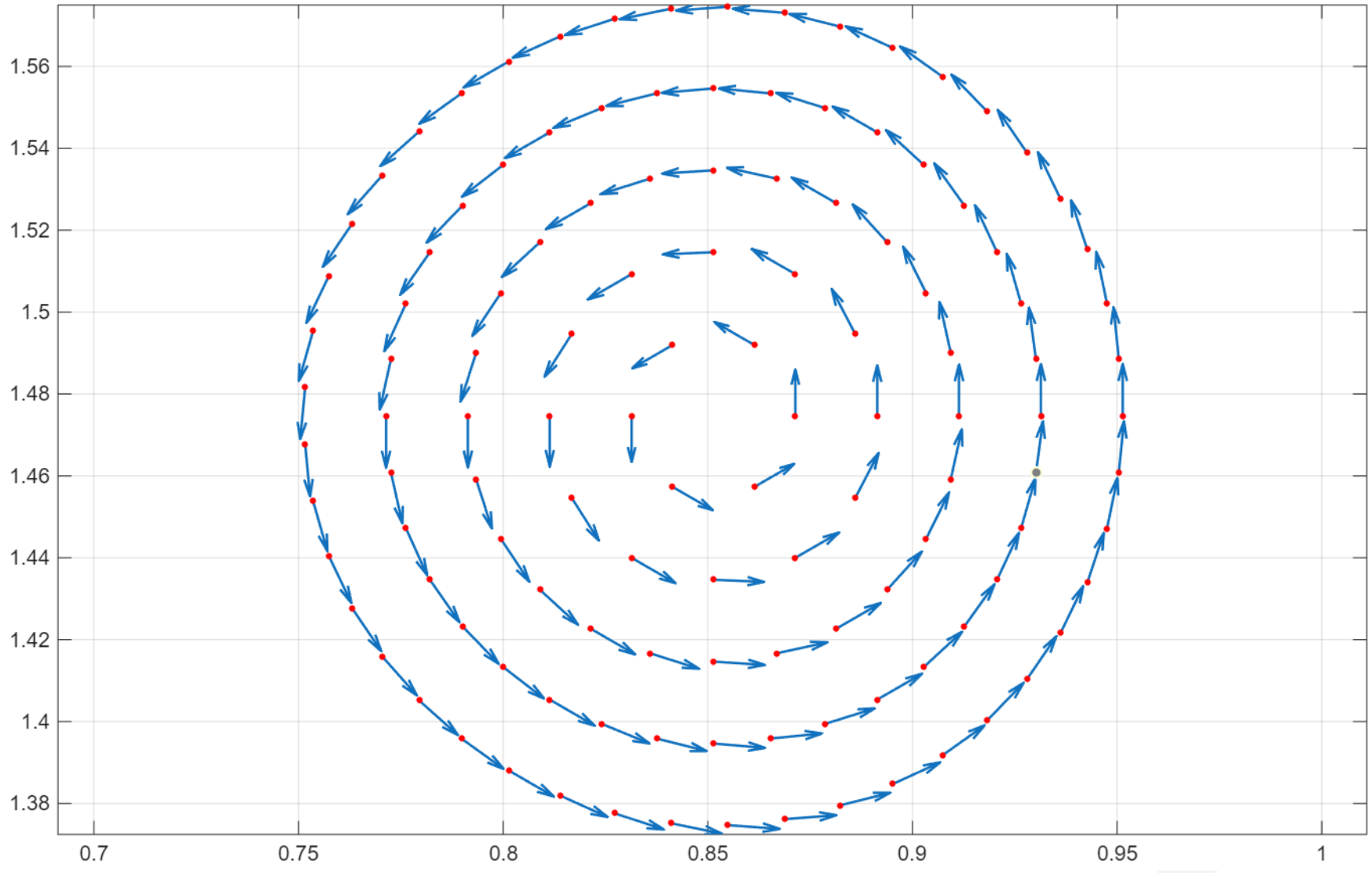
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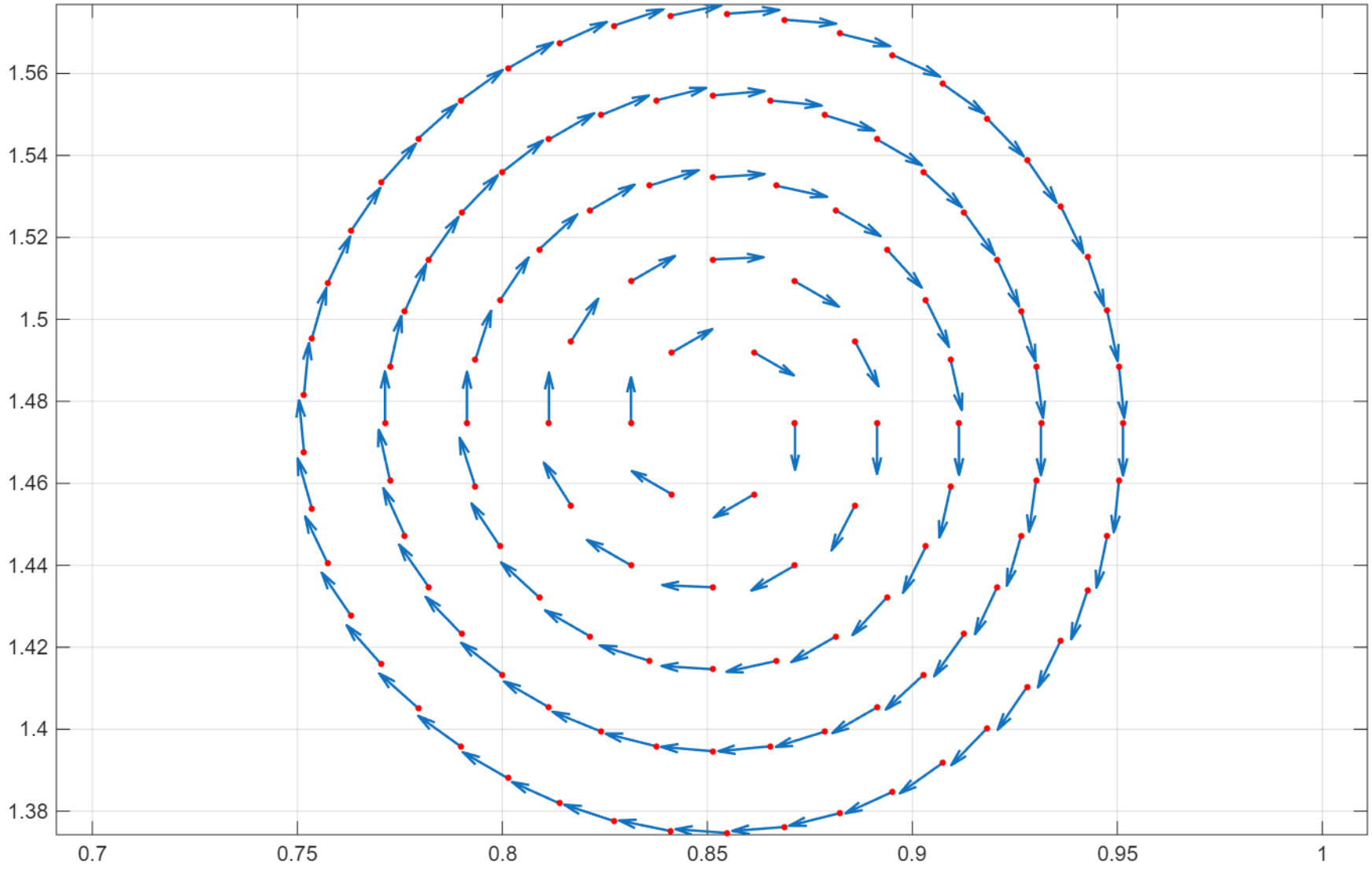
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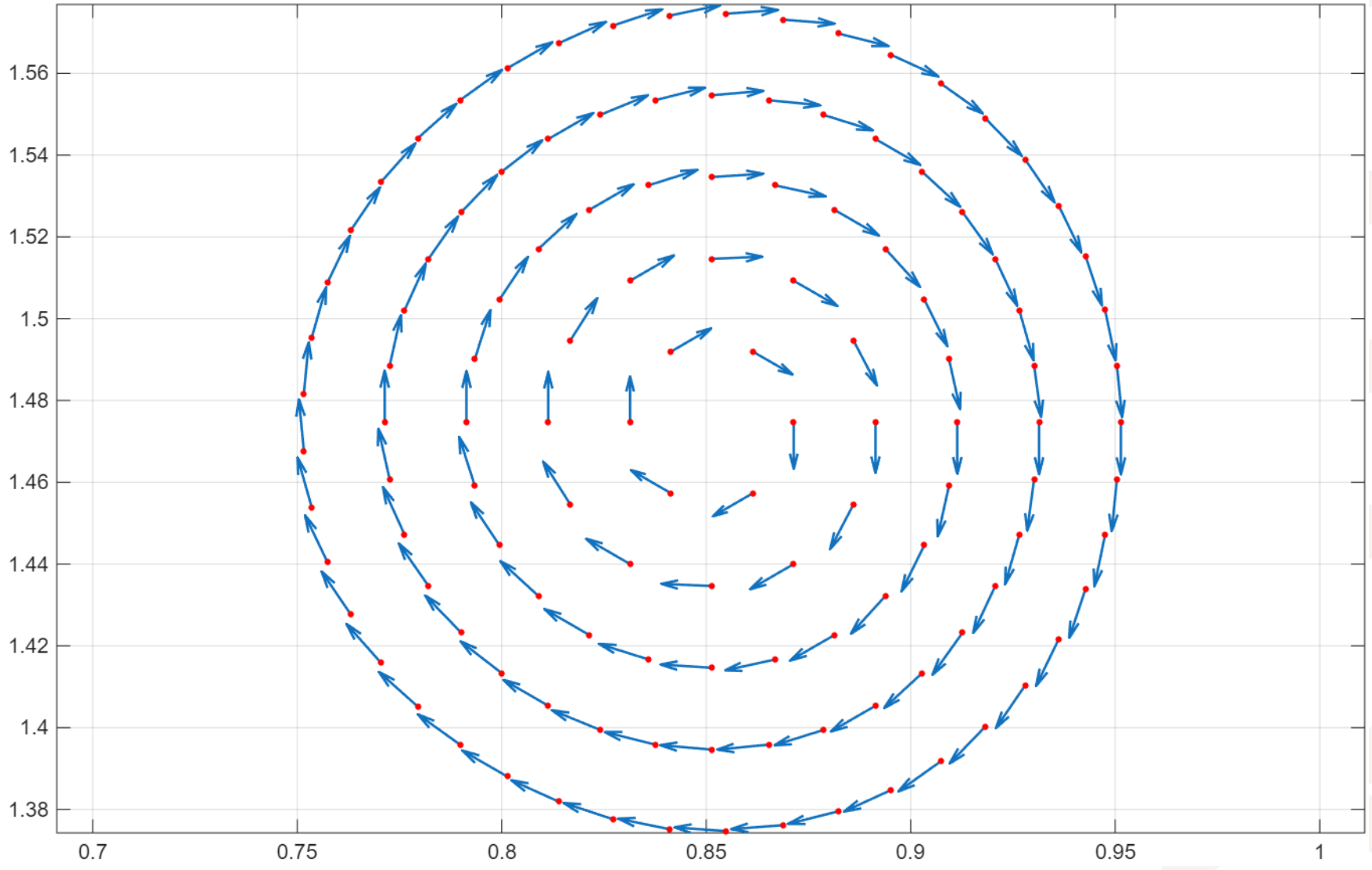
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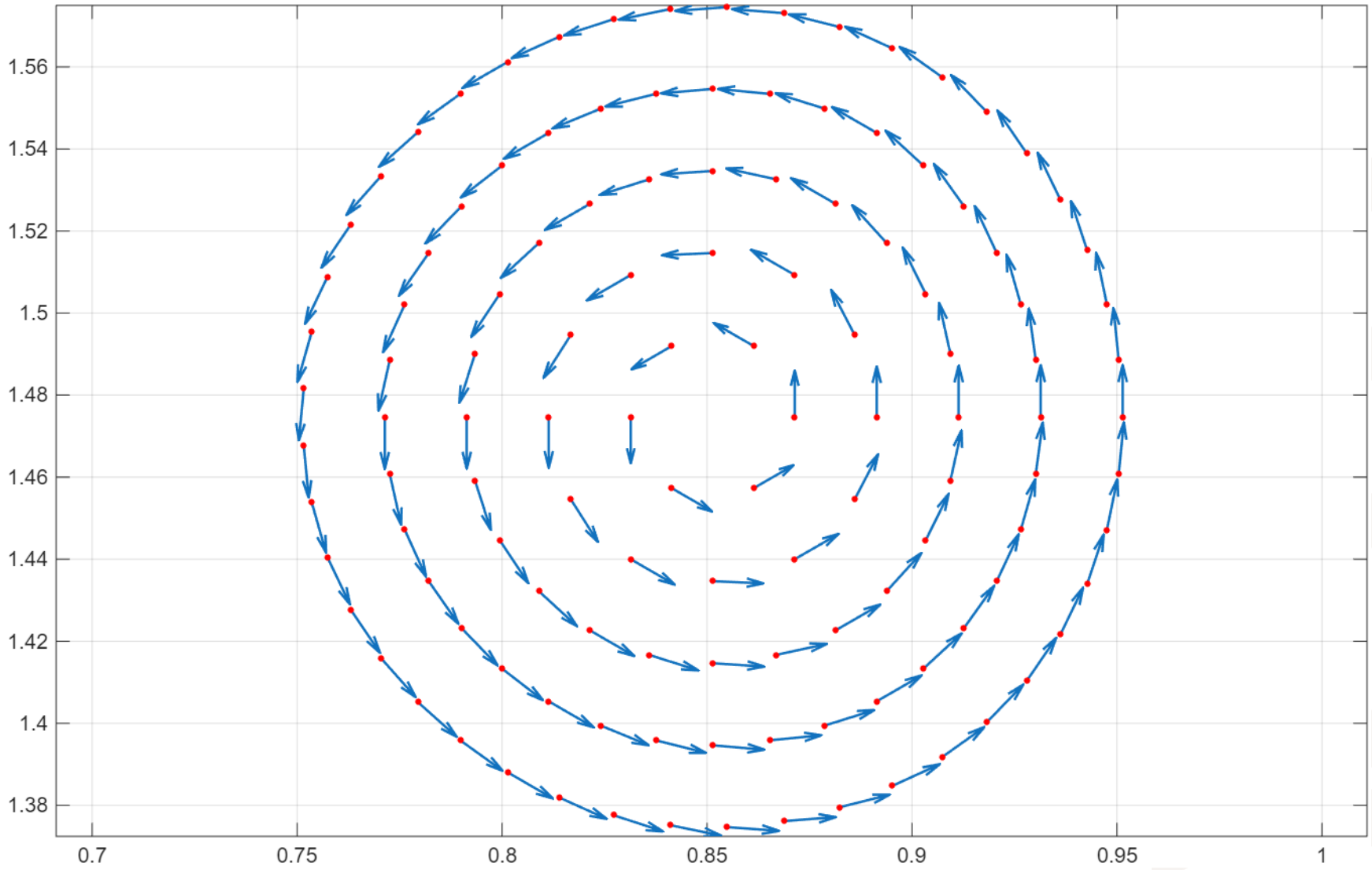
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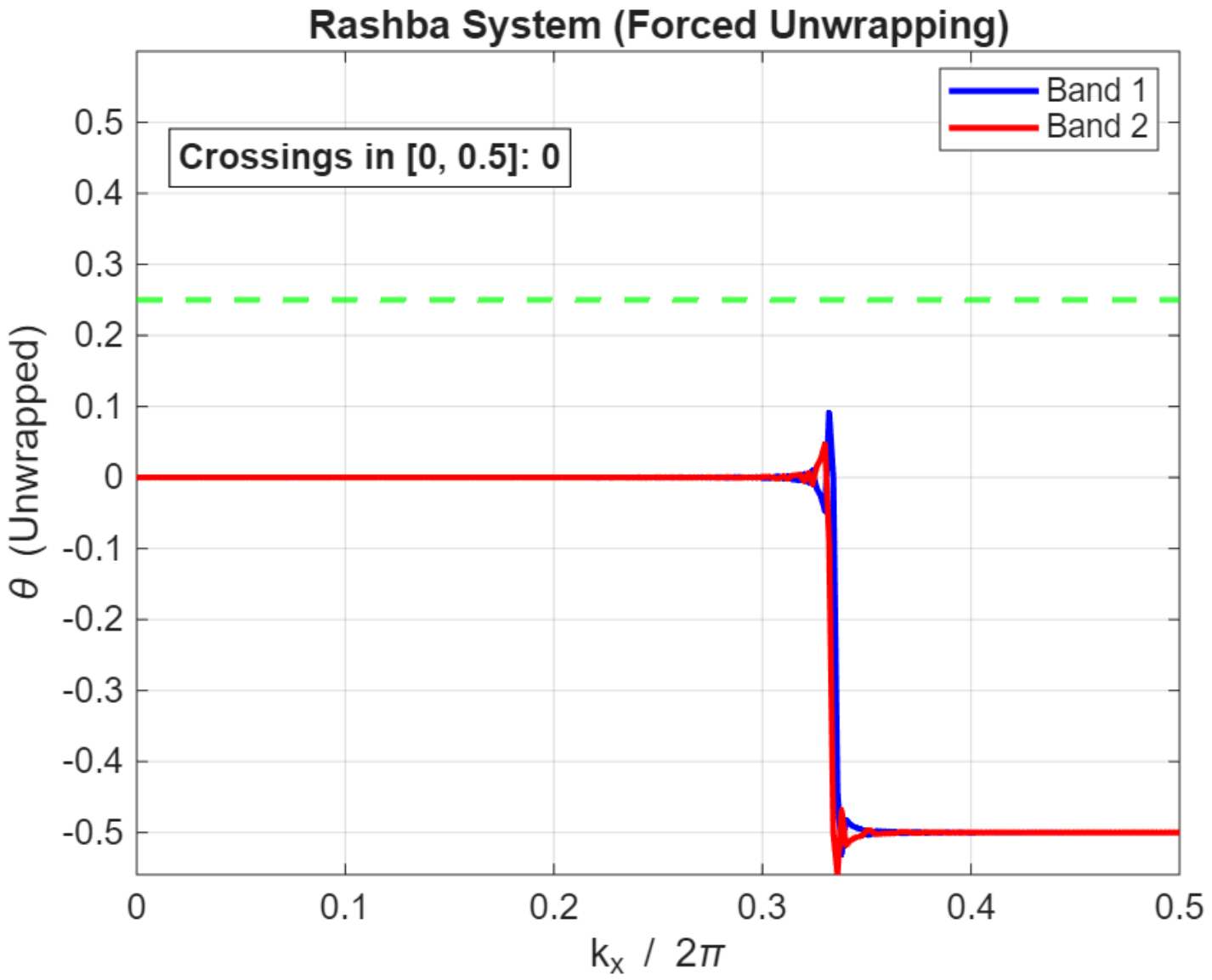
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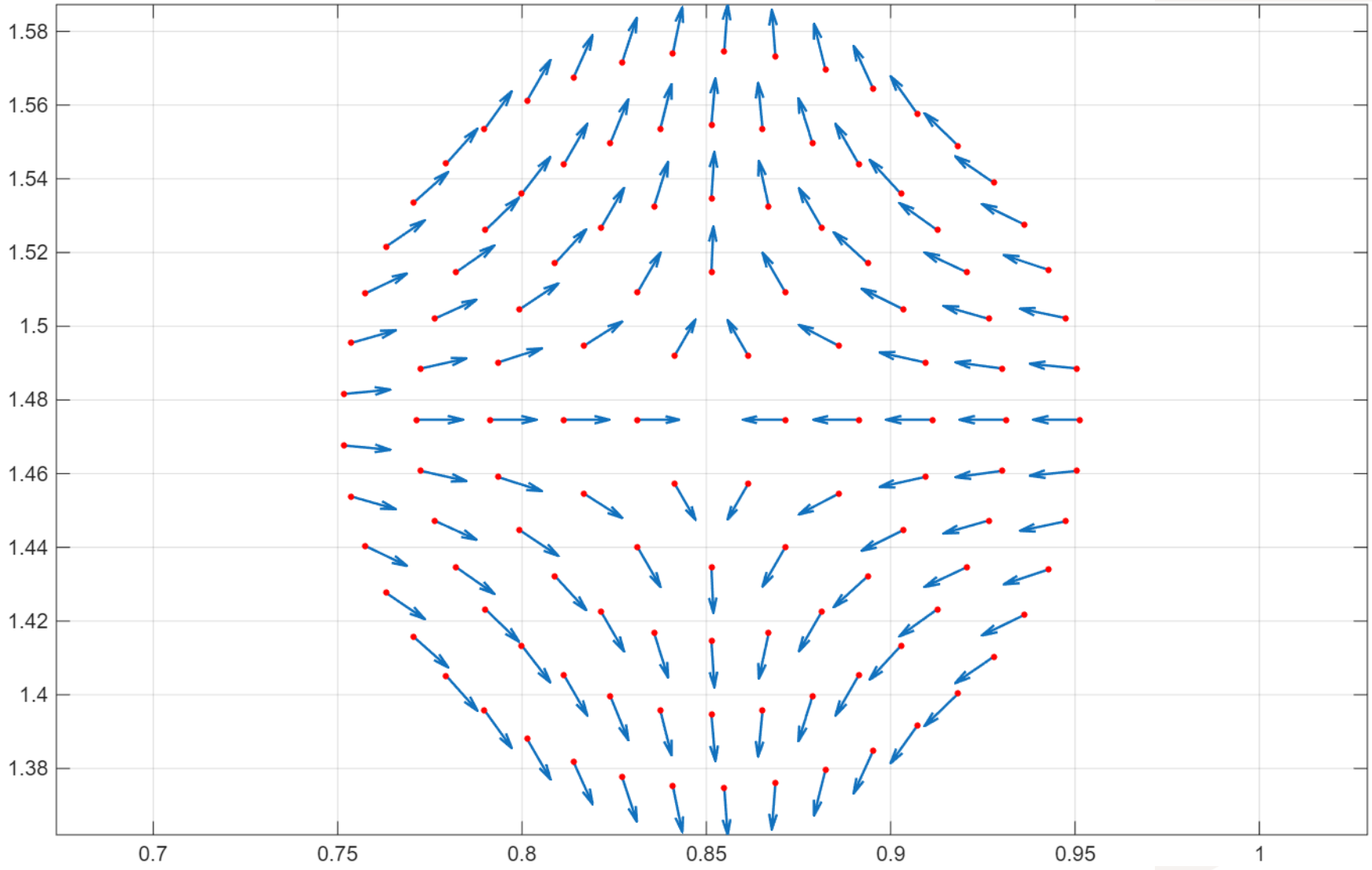
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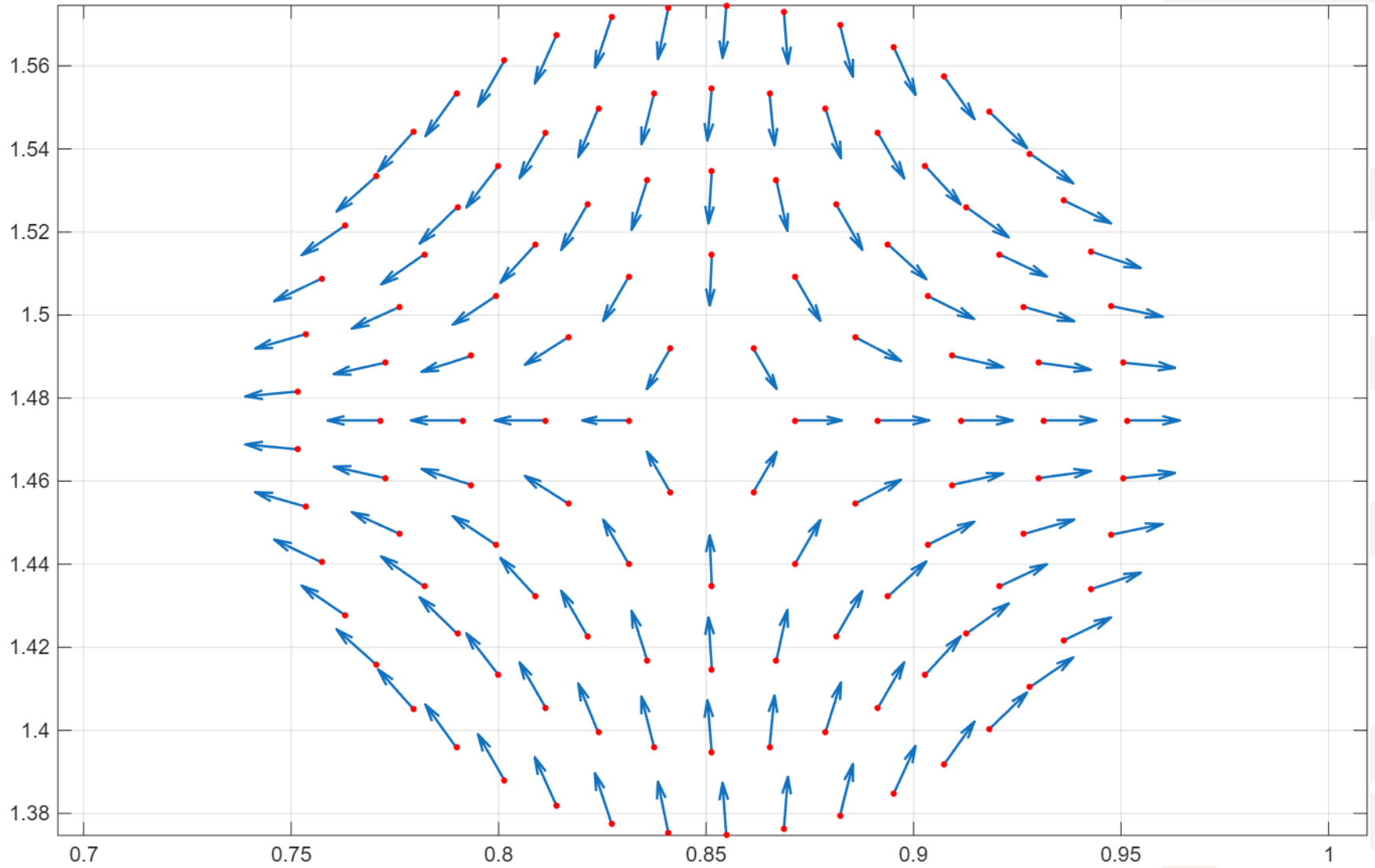
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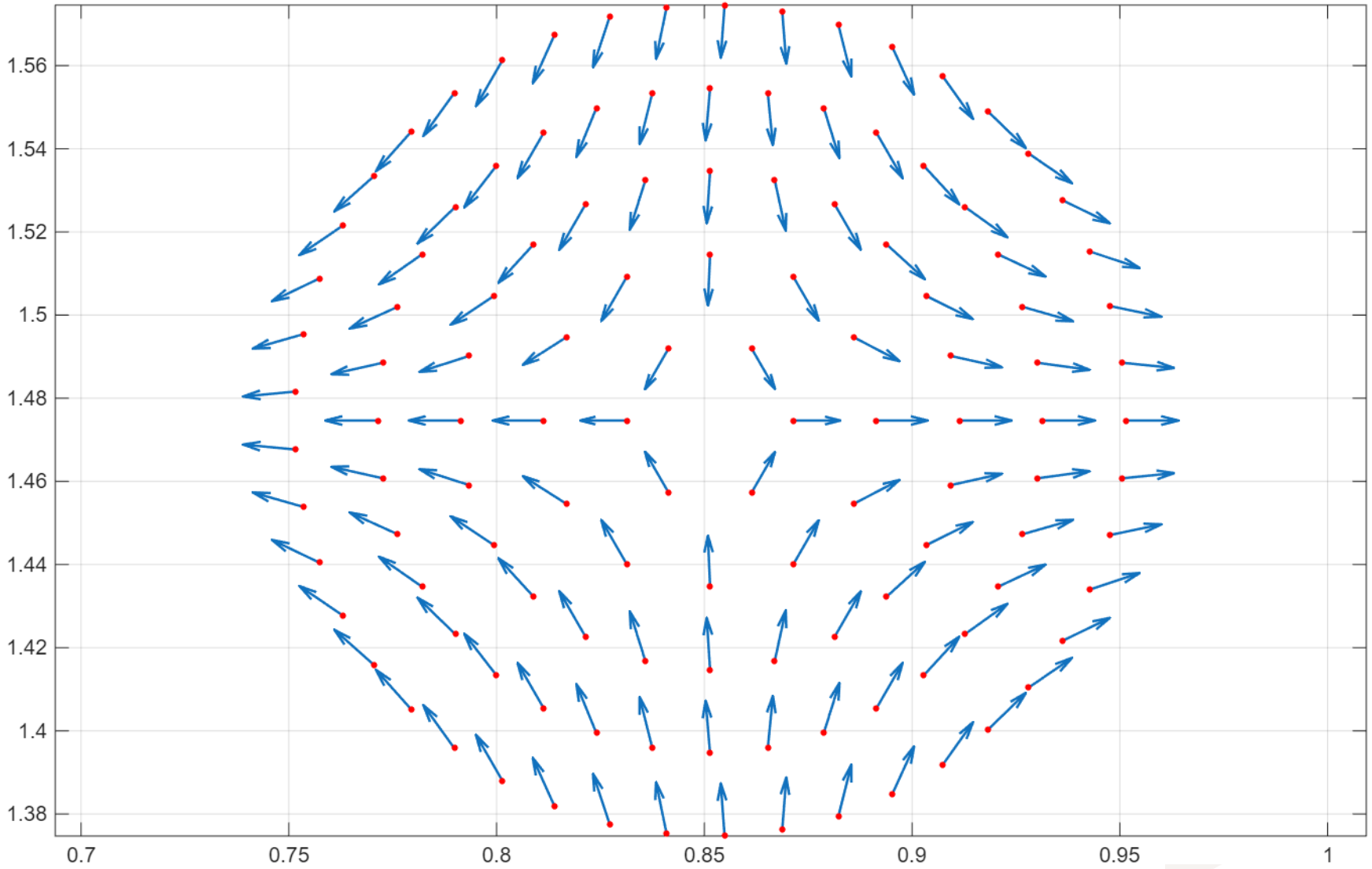
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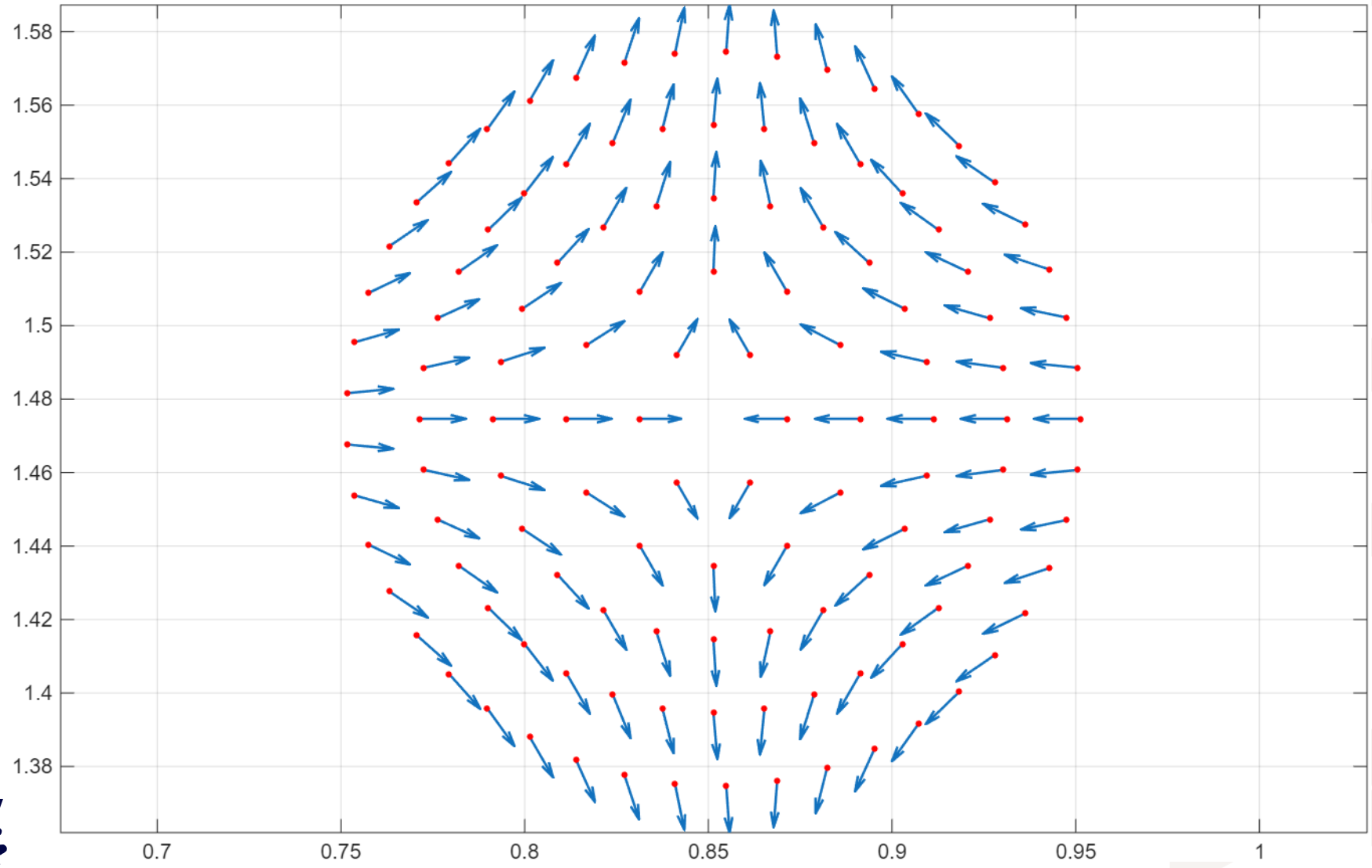
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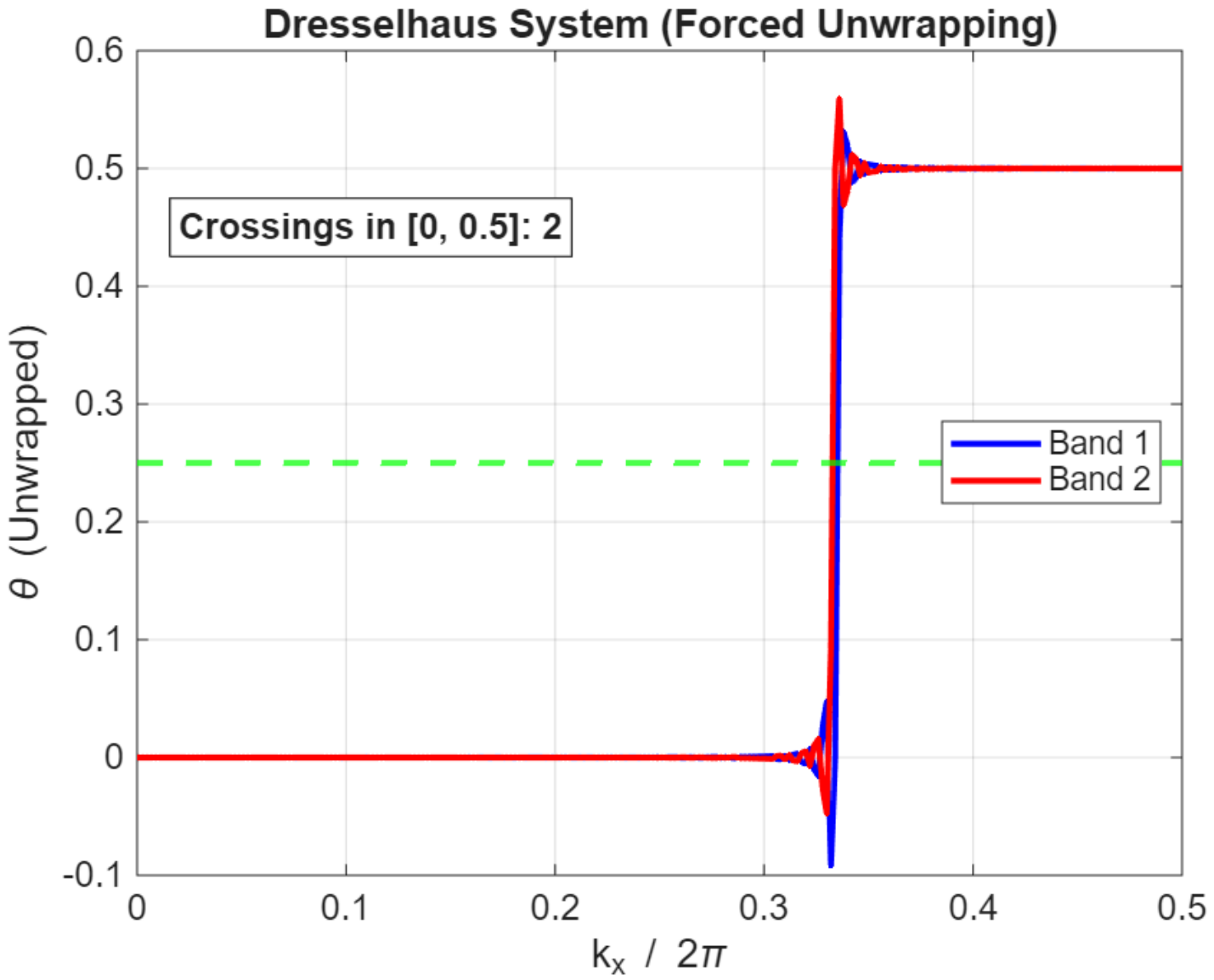
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That's all for this session

Thank you for your attention and great participation